

Role of Artificial Intelligence in the Management of Food Waste

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Abstract— Food waste is caused by a complex collection of interconnected behavior at both the supplier and customer levels. Computational and mathematical models provide various methods for simulating, diagnosing, and predicting various aspects of the dynamic food waste generation and prevention system. This paper describes three modeling methods that have been used to analyze food waste in the past. Bayesian networks and machine learning algorithms are applied to help determine how much food is discarded at the household level. Agent-Based Simulation was used to gain insight into how innovation and adoption of a particular technology can help minimize retail food waste. The first BN-ABM integrated model assesses consumer food waste levels affected by particular features and aspects, resulting in the model reaching equilibrium. Proofing that there is a need for policy interventions, including training, economic incentives, and campaigns, to obtain the resulting model transformation. These interventions are ready to be assessed by the model, but further study is needed to understand the effects of enforcing these structures on the accuracy of the BN-ABM predictive model. The second ABM model aims to determine the factors that establish the adoption of food waste reduction technology at the retail level, which is influenced by particular but not limited to economic factors, including a strong network between retailers and consumer's awareness regarding food waste reduction technology. These findings can study the effects of policy intervention regarding food waste reduction at the retail and consumer level. The third ABM model employed a general food chain network model consisting of consumers, traders, and producers to simulate the dynamic change of product flow between agents and to be able to assess the effect of agent's behavioral contrast. Resilience is measured by the ability to deal with shocks, and efficiency is the share of total food manages to be dispatched to consumers. At first glance, the simulation results seem to show a system trade-off between efficiency and resilience. Network chain structures with higher efficiency displayed more sensitivity to shocks, while networks with less efficiency show more resilience. However, there seem to be modifications in the results when applying several trading interactions and shock types. Resiliency and efficiency are affected by social aspects (trust and preference) in trading interaction between agents. An essential aspect of resilience is the agent's ability to switch links (trading partners) which shows the capability of reorganization. Insights regarding the research can be applicable when considering real-life food chain and structural reorganization to increase resilience and efficiency in meeting national food security goals.

Keywords—food waste, Bayesian Network, Agent-Based Modelling, machine learning, waste reduction, waste generation

I. INTRODUCTION

According to the findings in current studies, the total amount of food loss or food waste produced for human consumption equals about 1.3 billion tons, a third of the total production. It means that a significant portion of the energy used in food production is waste and that greenhouse gas emissions are increased [1]. Waste produces greenhouse gas emissions that are detrimental to the environment. The food industry, in particular, is culpable for 34% of global greenhouse gas emissions, with 6% coming from never consumed, wasted food [2]. This food waste emits methane gas when it decomposes in landfills during anaerobic(oxygen-free) conditions, where methane has 25-34 times more potent than CO₂ in global warming potential[3].

It is estimated that over two billion people do not hold access to adequate, safe, healthy, and nutritious food. That is when we consider the combination of the total number of people worldwide who are affected by moderate levels of food insecurity and those who suffer from severe hunger [4]. The hunger crisis becomes more severely damaged due to the COVID-19 pandemic. It is estimated that COVID-related hunger will kill about 12,000 people daily, possibly more than the number of people killed by the actual disease [5]. All of the squandered nutritious food could be used to feed those in need, helping to boost food security. If food waste is reduced, there would be more available food for consumption without raising the production level in farms [6]. There would be enough food to feed all of the world's undernourished people if only one-fourth of the thrown away food could be saved and preserved [7].

In retrospect to food waste, a general definition of food loss is the food that is lost/destroyed during any of the four processes of the supply chain, that is, concerning the production methods, processing technology, retail handling, and consumer activity [8]. In terms of origin and scope, the term 'food waste' differs from 'food loss' but, often than not, is used interchangeably. Food 'waste' is readily defined to have occurred during the consumer and retail stages. In contrast, food 'loss' occurs early in the supply chain, from harvest to distribution, making it unfit for human consumption [9].

Food is 'wasted' to a large degree at the consumption stage in the middle- and high-income countries, meaning it is discarded even though it is still fit for human consumption. More food is 'loss' in developing countries due to a lack of facilities and access to appropriate technology in the harvest and distribution stages [10]. The world's population is estimated to surpass 10.5 billion people by 2050. Food production must increase by at least 70% by 2050 to keep up with population growth. It will necessitate a massive increase in food production. However, land scarcity, a declining water level, weather fluctuations, and a rise in agricultural operating costs have made this difficult. Food losses during processing, post-harvest operations, and food waste during consumption may help narrow the gap between supply and demand.[11]

In general, food 'waste' is an extensive subject that has been explored in recent literature from various perspectives. Several experimental attempts and research have been made to determine and evaluate the socio-economic behavior linked to food waste drivers. These attempts were made to provide the correct understanding, interpretation, and elimination of food waste. The subject of this research includes the identification of the drivers of household-level food waste generation using Agent-Based Modelling and Bayesian Networks. Proper identification of them may aid in the development of successful food waste prevention and reduction policies. This study also aims to look into the essential aspects that affect retail level adoption of technology aimed to reduce food waste using Agent-Based Modelling (ABM).

II. METHODS

Drivers of Food Waste Generation at Household Level

This paper will use a combination of Bayesian network (BN) and agent-based modeling (ABM) to classify the household-level food waste drivers. Bayesian Networks (BN) can integrate complexity and ambiguity in the structure of the model. However, BNs are less successful at integrating behavioral variables (such as actor's interactions and individual actors' specific biases) and interaction across time between actors or variables. Agent-Based Models (ABM) are more suitable for this type of data.

The Bayesian network and ABM integration in the consumer model produces a simulation on the impact of social experiences and behavioral influences on the changes in an individual's opinions and behaviors about food waste, and thus on household food waste production. The model of food waste was developed as part of 2017 REFRESH by van Herpen, van Triip, and Van Geffen, in which a REFRESH-developed questionnaire is used as the source of data for the model calibration. The Food Waste model structure is developed by Van Herpen, van Triip, and van Geffen as part of REFRESH (2017). The REFRESH-developed questionnaire attempted to measure food waste by evaluating a collection of fixed characteristics and food-related behaviors in consumer household samples from countries such as Spain, Germany, Hungary, and the Netherlands.

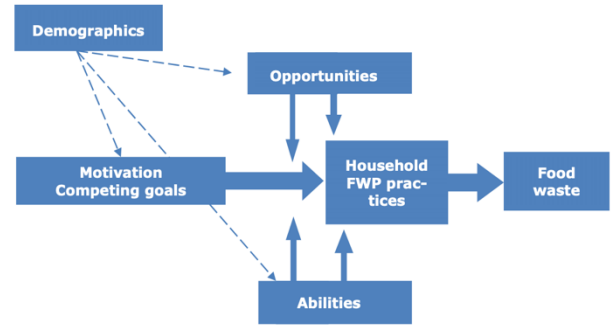


Figure 1. WP1 REFRESH Food Waste Model. Adapted from Grainger et al. (2019), p.19.

Six features can be categorized as 1. Socio-demographics (size of household, respondent's age and gender, income group, level of education, municipality size, and country); 2. Motivations (awareness of the harmful effects of food waste, attitudes against food waste); 3. Competing food goals (health, price, taste, and convenience in preparation time); 4. Food Waste Prevention (FWP) activities in households (planning food shopping and usage, avoiding impulse purchases, keeping track of what is in stock, cooking specifically, preserving and using a plate and pan leftovers); 5. Opportunities (product availability, shop accessibility, room and storage equipment availability, and the likelihood of unexpected events); 6. Abilities/skills (perceived difficulty with accurate preparation, determining food safety, cooking, and understanding shelf life and how to extend them); 7. Personality characteristics (getting involved in food preparation, financial control of food, parent control).

Figure 1 represents the correlation between these features in the Food Waste Model developed by REFRESH WP1.

Simulated populations are produced in the first step, using REFRESH questionnaire customer data. The Agent-Based Model then evolves as follows:

1. Each consumer agent will choose one of the six competing goals related to purchasing food;
2. For each consumer agent, another agent(friends, family, or neighbor) inside his/her networking circle whose averaged opinions/belief on the six competing goals is closest to the existing threshold will be analyzed ;
3. The consumer agent adjusts their opinion/belief on the chosen competing goals by averaging their opinion with the average opinion of their friends/family/neighbors, using weights that correspond to the chosen goal.
4. As a result of empirically found statistical associations between opinions, the agent's opinions on all other objectives change accordingly.
5. Changes to the agent's attitude toward food waste are calculated by averaging an agent's weights (influenceability) with the average weight(influenceability) of their neighbors;
6. Changes to the agent's perception towards the effects of their food waste are averaged with the average weights (influenceability) of their neighbors;
7. Each agent's neighbor's average behavior towards food waste is measured by the level of "importance" that each agent feels about the competing goals of "waste," this is used to measure what others think we do (injunctive of social norms);
8. The

The relationship between adoption of food waste-reduction innovation by retailers and food waste levels.

Small stores, discounts, and large-scale businesses are all represented as various types of retail agents. Each retail agent must only adopt one of the two technologies: a standard that all retail agents initially adopt. A massive amount of food waste will be produced or adopted an advanced technology that reduces consumer food waste at home or produced in stores. Retail agents use a utility feature that includes three key elements to determine whether or not to implement the low-waste innovation: (1) the profit made, which is determined by selling rates, research, cost of development, and the percentage of food lost after a purchase by customer or food lost in stores; (2) environmental issues, as well as reputational concerns associated with pro-environmental behavior; (3) decisions made by other retail agents.

The retail model's intra-step measures are as follows:

technology; 3. Customers who purchased from a retail agent switched their technology adoption to the same retailer agent; 4. Consumers learn about the low waste technology, and the information may spread among customers through words of mouth or from retailer advertising; 5. Based on the previously mentioned parameter, such as utility and knowledge status, many customers with common attributes agree to switch retail agents; 6. There is a recalculation of the new market shares of each retail agent, and then a new phase will begin.

A generic network of food supply chains made up of producers/farmers, traders/distributors, and consumers can be modeled using ABM. It is done to explain the dynamically evolving disaggregated crop item flows between them, alongside understanding their behavior. For trading between producers and traders, or traders with the consumer, each agent has implied personal goals. The trading link can be formed randomly or with applied personal goals (i.e., best price and trading history). Under various types of stylized shocks: 1. Consumer shock (preference change) 2. Trader shock (stock loss) 3. Producer shock (failed harvest) also considers the different interactions between agents alongside the network structure. We measure efficiency (amount of total food managed to be dispatched to consumers) and resilience (ability to cope with shocks) by comparing the variable changes to the percentage of the overall explicit goal of having all consumers meet their food needs.

The integrated BN-ABM model helps us incorporate interpersonal dynamics into the study of a policy intervention while still using the explanatory capacity of a Bayesian Network model to measure conditional probabilities in the presence of many variables. Here are the studied results :

[illegible]

Figure 2. Semi-structured Bayesian Network used to estimate agents' food waste levels in the Integrated consumer model. It was adapted from Grainger et al. (2019), p.15.

After the intra-step in retail ABM for all agents is completed, these new values of consumer motivation and competing goals are then submitted to the customer BN as seen in Figure 2. The dark green node indicates consumer motivation, and the light green node indicates competing goals. The BN then calculates the probability of each five categories that a consumer food waste falls into (low-high). Next, from each agent's probability distribution, a particular value of food waste is derived. In some cases, when the BN cannot construct a distribution probability for some consumer agent, the model assumes that the probability amounts to 20% for each of the five food waste level classes. Using the estimated distribution probability, the model then calculates the agent's total amount of food waste. They were seen in Figure 1 depicted by the red node indicating food waste output in grams. This new food waste level will become the parameter of a new time phase in the ABM model.

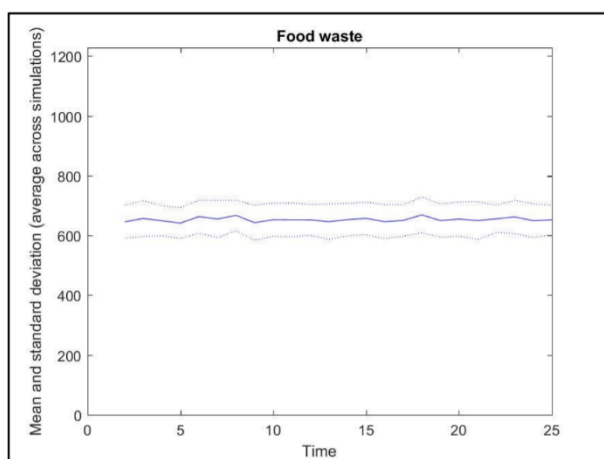


Figure 3. The mean and standard deviation of food waste (grams) across simulated populations at Each time step. It was adapted from Grainger et al. (2019), p.25.

Initial simulations in Figure 3 demonstrates that the model is in balance/equilibrium. In the short term, when the population structure does not change, and there is no policy adoption to minimize food wastage or related shocks (scandals of food safety, etc.), the equilibrium model is anticipated. That is because the influence of motivational shifts on food waste is minimal. This is most likely due to the consumers' social desirability bias, which distorts the data from the questionnaire calibration and prevents the identification of true motives. Since the model is basically in equilibrium, it is plausible to concentrate on the slight impact made by policy adoption and policy decision in cross-sectional and dynamic terms.

Following the adaptation of the theoretical framework, 2 of the features mentioned above, abilities and opportunities, moderate/control the correlation between competing goals/motivations and actions that causes food waste, causing the impact of the competing goals/motivations and action to not be in linearity. The BN structure established a strong prioritization on motivations (awareness and attitude towards the effect of food waste) and its impact on food waste behavior, which is implausible because the data on the WP1 survey revealed a significant impact of social norms on the

level of food waste. To maintain a more robust structure, 1,000 neural networks were built from the WP1 dataset to assess the relative "importance" of all the variables by summing up the weights of each input variable and having the total amount of food wasted in grams as output. After machine learning the Bayesian Network and only retaining the connected and important nodes which entirely relates to "action_leftovers"; "The leftovers I have will be stored and eaten," "If I served too much food, the leftovers would be stored," an increase of prediction accuracy from 47% to 69% has been found in the model.

The relationship between adoption of food waste-reduction innovation by retailers and food waste levels.

The final outputs are the total food waste produced in the industry and the retailer's market share who adopted the technology of low-food waste (which can be seen on REFRESH Deliverable 4.5). Some examples of innovations that retail stores can adopt include; A storage device that extends vegetables and fruits shelf life and a type of bag that minimizes the exposure of fresh produce to environmental conditions.

The simulations performed using ABM show that retailer's adoption of new low waste technology is a dynamic mechanism driven by a range of factors other than solely economic factors.

The strength of retailer networks, especially among small and medium businesses, appears to have the most significant impact on retailers' decisions about whether or not to implement food waste reduction innovations. Despite the differences in their grocery market systems, the position of retail networks is essential in both Italy and the Netherlands.

An equally important factor that plays a role in low-waste technology adoption is the level of customer knowledge about whether or not low food waste technology exists. The joined impact of high customer knowledge and the existence of substantial networking between retailers, as predicted, contributes to a broader acceptance of reduction technologies regarding food waste. Retail businesses with high interaction between retail agents and excellent knowledge among customers, in particular, have a high rate of adoption of technologies that reduce food waste.

Each invention makes a small but significant contribution to food waste reduction in terms of percentage but has a significant effect on absolute amounts. Specifically, the evaluated simulation results show that the food waste at the retail level in the Dutch case ranges from 2 to 3.6% and from 0.8 to 2.1% in the Italian case.

ABM model to assess the efficiency and resilience of food supply chain

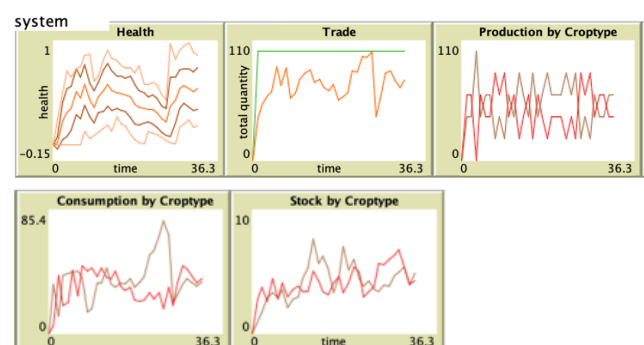


Figure 4. A modeling framework before the consumer-type shock is applied

The above modeling framework (adapted from Van Voorn G et al. (2020)), In the experimental design, simulations are produced through different settings that made up the model. As seen in Figure 4, we determine the resilience among different networks and compare them through these simulations. The above modeling output is just one of the many that we use to produce studied observations to model the impact of different factors. When applied a consumer shock factor over 15 years, the modeling output is generated :

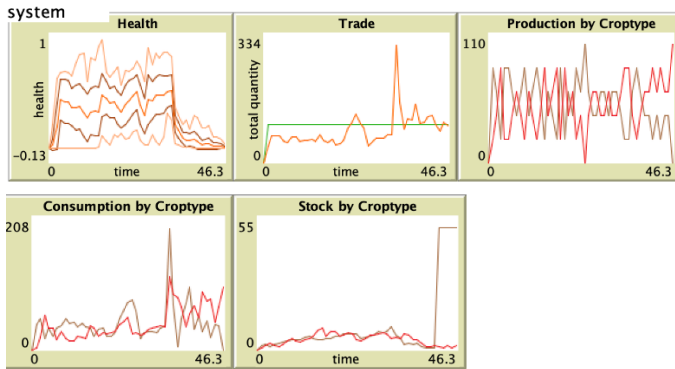


Figure 5. A modeling framework after consumer-type shock is applied

The first point to note is that shocks to consumption have a much more significant effect than shocks to production or trading. The consumption form of shock is linked to shifts in the preferences of consumers for different crops, as well as the total time required by traders and producers to react. To put it another way, the effect is a time lag because farmers must change their products, and traders must stockpile the new crop. Tinier networks have a wider variety regarding effects than broader networks regarding shocks at the trading stage. It is understandable since smaller networks have fewer agents and thus exhibits more noise, especially evident as shocks operate at the intermediate network level, where there are more connections between producers, traders, and consumers.

The following examination shows that weighted(based on best price) interactions have more influence than random interaction or preference (based on best price and past transaction). This is caused by the fact that agents who proceed with weighted interactions will cause the network to be more productive (considering crops transfer faster from producer to consumer) than having agents with forms of other interaction. However, this type of network has a lower level of resilience. Agents in weighted interactions look for the best offer based on the optimization of economic utility. Therefore, dietary requirements are more effectively fulfilled for consumers. There is a disadvantage in which shocks take on a higher effect in weighted interactions than with preference or random interaction. In random encounters, links between the not optimal agents or even "negative" are often randomly formed and broken. Therefore, it causes efficiency and the impact to be lower because alternatives will rapidly replace broken connections due to randomization. Less practical economic utility

happens in interactions based on best price and past transaction (preference-based) since many possible connections can be missed due to the preference. Agents, on the other hand, favor trusted partners over less expensive ones, which decreases performance. As a result, the shock's effect is also reduced.

The subsequent examination is that the variety of effects depends on the network type. Except for the shock at the trading stage, the effect is most significant in the network where there is more producer and consumer than traders, proving to be the most powerful network form. This is due to the lack of traders and options to choose from, so fast trades of crops happen between supplier, trader, and consumer. Meanwhile, networks with more traders than producers and consumers have the tiniest effect but the highest amount of stock. These networks seem to be inefficient since there are more traders than producers and customers. The advantage of this inefficiency is that it makes the agents less vulnerable to shocks.

Furthermore, there seems to be an unrealistic aspect to the agents depicted in the model. They act according to the optimal economic principles and exhibit no different rationality, strategic thinking, social dynamics, adaptation, and learning ability. Network resilience is built upon agents' action in reconstructing their trading network according to economic principles, which is not always realistic agent behavior. Another simplification is that agents build only one link each step of the Simulation, although realistically, they could have numerous links adapted simultaneously. The agent-based model has also yet to be validated using real-world data. Information regarding the area, region, income categories, occupation groups, household size, education, age, gender, and economic activity can be utilized to calibrate the model at different points in time. Moreover, other crucial elements remain unaccounted for, including information on who communicates/trades with whom and why and the process in decision-making.

IV. CONCLUSION

Despite their complexity, Bayesian Networks and Agent-Based Modelling are AI-implementations that might help with data analysis and decision making. Both modeling methods, however, have drawbacks. It is important to note that BN and ABM have made it possible to understand the interdependencies between certain variables, although not their causality or mechanism. It is essential to understand why country-level and age disparities exist to develop improved food waste policy approaches, which will need further study using various methods.

Agent-Based Modelling is suitable in studies of food waste generation phenomenon, resulting from accumulated individual decisions and the cause of those decisions that are interrelated and complex. Two ABMs were developed as part of the EU Horizon 2020 REFRESH (Resource Effective Food and drink for the Entire Supply Chain) project to investigate the relationship between food waste generation and innovation adoption in the retail sector. Resilience and efficiency assessment of food supply chain is also developed using Agent-Based Model (ABM) that includes explicitly

rules of social behavior and decision-making, how agents interact inside networks, and changes in the population, while also allowing for system dynamics.

Resilience is measured by the agents' ability to change trading partners, also known as their ability to reorganize. These findings may help reorganize the food chain system of the natural world to improve real network resilience and achieve potential nutrition and food protection goals.

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